PRACTICE QUESTIONS CONDITIONAL PROBABILITY

EXERCISE – 2.7

1. A card is picked at random from a pack of 20 cards numbered 1,2,3,…..,20. Given that the card shows an even number, find the probability that it is a multiple of 4.
$(Ans: 0.5)$

2. A number is picked at random from the digits 1,2,….,9. Given that the number is a multiple of 3, find the probability that the number is
(a) even (b) a multiple of 4
$(Ans: 1/3, 0)$

3. In a large group of people it is known that 10% have a hot breakfast, 20% have a hot lunch and 25% have a hot breakfast or a hot lunch. Find the probability that a person chosen at random from this group.
(a) has a hot breakfast and a hot lunch
(b) has a hot lunch given that the person had a hot breakfast.
$(Ans: 0.05, 0.5)$

4. Two tetrahedral dice with faces labeled 1,2,3,4 are thrown and the number on which each lands is noted. The score is the sum of these two numbers. Find the probability that
(a) the score is even given that at least one die lands on a three
(b) at least one die lands on a three given that the score is even
$(Ans: 3/7, 3/8)$

5. In a group of 100 people, 40 own a cat, 25 own a dog and 15 own a cat and a dog. Find the probability that a person chosen at random
(a) owns a cat given that he owns a dog
(b) does not own a cat given that he owns a dog
(c) does not own a cat given that he does not own a dog
$(Ans: 3/5, 2/5, 2/3)$

6. A box contains 3 white and 2 black balls. Two balls are drawn in succession. Find the probability that both balls drawn are black when the balls are not replaced after being drawn.
$(Ans: 1/10)$

7. A box contains 15 items, 4 of which are defective and 11 are good. Two items are selected. What is the probability that the first is good and the second defective when the balls are not replaced after being drawn.
$(Ans: 22/105)$

8. Two cards are dealt from a pack of ordinary playing cards. Find the probability that the second card dealt is a heart.
$(Ans: 1/4)$

9. A fruit basket contains 25 apples and oranges, of which 20 are apples. If two fruits are randomly picked in sequence, what is the probability that both the fruits are apples?
$(Ans: 0.633)$

10. A bag contains four red counters and six black counters. A counter is picked at random from the bag and not replaced. A second counter is then picked. Find the probability that
(a) the second counter is red given that that the first counter is red
(b) both counters are red **(c)** the counters are of different colors
$(Ans: 1/3,\ 2/15, 8/15)$

HAZBER SAMSON

PRACTICE QUESTIONS COMBINED EVENTS

EXERCISE – 2.8

1. Events A and B are such that $P(A|B)=\frac{2}{5}$, $P(B)=\frac{1}{4}$, $P(A)=\frac{1}{3}$ Find

(a) $P(A\cap B)$

(b) $P(B|A)$

(c) $P(A^c|B^c)$

$(Ans: 1/10,\ 3/10,\ 31/45)$

2. Events C and D are such that $P(C)=\frac{4}{7}$, $P(C\cap D')=\frac{1}{3}$, $P(C|D)=\frac{5}{14}$

Find

(a) $P(C\cap D)$

(b) $P(D)$

(c) $P(D|C)$

$(Ans: 5/21,\ 2/3,\ 5/12)$

3. Two Events A and B are such that, $P(A)=8/15$, $P(B)=1/3$, $P(A|B)=1/5$

Calculate the probabilities that

(a) both events occur

(b) only one of the two events occur

(c) neither event occurs

$(Ans: 1/15, 11/15, 1/5)$

4. Two events A and B are such that $P(A)=0.2$, $P(B)=0.3$, $P(A\cup B)=0.4$. Find the following probabilities

$(a)\ P(A\cap B)$

$(b)\ P(A^c\cup B^c)$

$(c)\ P(A^c\cap B^c)$

$(d)\ P(A^c|B)$

$(Ans: 0.1,\ 0.9,\ 0.6,\ 2/3)$

5. The Events A and B are such that, $P(A)=0.45$, $P(B)=0.35$, $P(A\cup B)=0.7$

Find

(a) $P(A\cap B)$

(b) $P(A|B)$

(c) $P(A\cap B')$

(d) $P(A^c|B)$

(e) $P(A^c|B^c)$

$(Ans: 1/10,\ 2/7,\ 7/20,\ 5/7,\ 6/13)$

HAZBER SAMSON

6. If A and B are independent events with $P(A)=0.5$ *and* $P(B)=0.2$, find the following probabilities

$(a)\ P(A\cup B)$ $(b)\ P(A^c\cap B^c)$ $(c)\ P(A^c\cup B^c)$

$(Ans: 3/5,\ 2/5,\ 9/10)$

7. A and B are two independent events such that $P(A)=0.2$ *and* $P(B)=0.15$ Evaluate the following probabilities.

(a) $P(A|B)$ (b) $P(A\cap B)$ (c) $P(A\cup B)$

$(Ans: 0.2, 0.03, 0.32)$

8. The probability that an event occurs is $P(A)=0.4$, B is an event independent of A and probability of the union of A and B is $P(A\cup B)=0.7$. Find

(a) $P(B)$ (b) $P(A\cap B)$ (c) $P(A|B)+P(B|A)$

$(Ans: 0.5, 0.2, 0.9)$

9. Events A and B are such that, $P(A)=0.4$ *and* $P(B)=0.25$. If A and B are independent events, find

(a) $P(A\cap B)$ (b) $P(A\cap B^c)$ (c) $P(A^c\cap B^c)$ (d) $P(A|B)$ (e) $P(A^c|B^c)$

$(Ans: .0.1, 0.3, 0.45, 0.4, 0.6)$

10. There are 90 applicants for a job with the news department of a television station. Some of them are college graduates and some are not; some of them have at least three years' experience and some have not, with the exact breakdown being

	College graduates	*Not college graduates*
At least three years' experience	18	9
Less than three years' experience	36	27

If the order in which the applicants are interviewed by the station manager is random, *G* is the event that the first applicant interviewed is a college graduate, and *T* is the event that the first applicant interviewed has at least three years' experience, determine each of the following probabilities directly from the entries and the row and column totals of the table:

(a) $P(G)$; **(b)** $P(T')$; **(c)** $P(G\cap T)$;

(d) $P(G'\cap T')$; **(e)** $P(T|G)$; **(f)** $P(G'|T')$.

$(Ans: 3/5,\ 7/10,\ 1/5, 3/10,\ 1/3, 3/7)$

HAZBER SAMSON